

Laplace Transforms

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21.1 INTRODUCTION

The knowledge of Laplace transforms has in recent years become an essential part of mathematical background required of engineers and scientists. This is because the transform methods provide an easy and effective means for the solution of many problems arising in engineering.

This subject originated from the operational methods applied by the English engineer Oliver Heaviside (1850–1925), to problems in electrical engineering. Unfortunately, Heaviside's treatment was unsystematic and lacked rigour, which was placed on sound mathematical footing by Bromwich and Carson during 1916–17. It was found that Heaviside's operational calculus is best introduced by means of a particular type of definite integrals called Laplace transforms.*

The method of Laplace transforms has the advantage of directly giving the solution of differential equations with given boundary values without the necessity of first finding the general solution and then evaluating from it the arbitrary constants. Moreover, the ready tables of Laplace transforms reduce the problem of solving differential equations to mere algebraic manipulation.

21.2 (1) DEFINITION

Let $f(t)$ be a function of t defined for all positive values of t . Then the **Laplace transforms** of $f(t)$, denoted by $L\{f(t)\}$ is defined by

$$L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt \quad \dots(1)$$

provided that the integral exists. s is a parameter which may be a real or complex number.

$L\{f(t)\}$ being clearly a function of s is briefly written as $\bar{f}(s)$ i.e., $L\{f(t)\} = \bar{f}(s)$,

which can also be written as $f(t) = L^{-1}\{\bar{f}(s)\}$.

Then $f(t)$ is called the **inverse Laplace transform** of $\bar{f}(s)$. The symbol L , which transforms $f(t)$ into $\bar{f}(s)$, is called the **Laplace transformation operator**.

*Pierre de Laplace (1749–1827) (See footnote p. 18) used such transforms, much earlier in 1799, while developing the theory of probability.

(2) Conditions for the existence

The Laplace transform of $f(t)$ i.e., $\int_0^\infty e^{-st} f(t) dt$ exists for $s > a$, if

(i) $f(t)$ is continuous

(iii) $\lim_{t \rightarrow \infty} \{e^{-at} f(t)\}$ is finite.

It should however, be noted that the above conditions are sufficient and not necessary.

For example, $L(1/\sqrt{t})$ exists, though $1/\sqrt{t}$ is infinite at $t = 0$.

21.3 TRANSFORMS OF ELEMENTARY FUNCTIONS

The direct application of the definition gives the following formulae :

$$(1) L(1) = \frac{1}{s} \quad (s > 0)$$

$$(2) L(t^n) = \frac{n!}{s^{n+1}}, \text{ when } n = 0, 1, 2, 3, \dots \quad \left[\text{Otherwise } \frac{\Gamma(n+1)}{s^{n+1}} \right]$$

$$(3) L(e^{at}) = \frac{1}{s-a} \quad (s > a)$$

$$(4) L(\sin at) = \frac{a}{s^2 + a^2} \quad (s > 0)$$

$$(5) L(\cos at) = \frac{s}{s^2 + a^2} \quad (s > 0)$$

$$(6) L(\sinh at) = \frac{a}{s^2 - a^2} \quad (s > |a|)$$

$$(7) L(\cosh at) = \frac{s}{s^2 - a^2} \quad (s > |a|)$$

Proofs. (1) $L(1) = \int_0^\infty e^{-st} \cdot 1 dt = \left[-\frac{e^{-st}}{s} \right]_0^\infty = \frac{1}{s}$ if $s > 0$.

$$(2) \quad L(t^n) = \int_0^\infty e^{-st} \cdot t^n dt = \int_0^\infty e^{-p} \cdot \left(\frac{p}{s}\right)^n \frac{dp}{s}, \text{ on putting } st = p$$

$$= \frac{1}{s^{n+1}} \int_0^\infty e^{-p} \cdot p^n dp = \frac{\Gamma(n+1)}{s^{n+1}}, \text{ if } n > -1 \text{ and } s > 0. \text{ [Page 302]}$$

In particular $L(t^{-1/2}) = \frac{\Gamma(1/2)}{s^{1/2}} = \frac{\sqrt{\pi}}{s}$; $L(t^{1/2}) = \frac{\Gamma(3/2)}{s^{3/2}} = \frac{\sqrt{\pi}}{2s^{3/2}}$

In n be a positive integer, $\Gamma(n+1) = n!$ [(v) p. 302],

therefore,

$$L(t^n) = n! / s^{n+1}.$$

$$(3) \quad L(e^{at}) = \int_0^\infty e^{-st} \cdot e^{at} dt = \int_0^\infty e^{-(s-a)t} dt = \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^\infty = \frac{1}{s-a}, \text{ if } s > a.$$

$$(4) \quad L(\sin at) = \int_0^\infty e^{-st} \sin at dt = \left[\frac{e^{-st}}{s^2 + a^2} (-s \sin at - a \cos at) \right]_0^\infty = \frac{a}{s^2 + a^2}$$

Similarly, the reader should prove (5) himself.

$$(6) \quad L(\sinh at) = \int_0^\infty e^{-st} \sinh at dt = \int_0^\infty e^{-st} \left(\frac{e^{at} - e^{-at}}{2} \right) dt$$

$$= \frac{1}{2} \left[\int_0^\infty e^{-(s-a)t} dt - \int_0^\infty e^{-(s+a)t} dt \right] = \frac{1}{2} \left[\frac{1}{s-a} - \frac{1}{s+a} \right] = \frac{a}{s^2 - a^2} \text{ for } s > |a|.$$

Similarly, the reader should prove (7) himself.

21.4 PROPERTIES OF LAPLACE TRANSFORMS

I. Linearity property. If a, b, c be any constants and f, g, h any functions of t , then

$$\mathbf{L} [af(t) + bg(t) - ch(t)] = a\mathbf{L} \{f(t)\} + b\mathbf{L} \{g(t)\} - c\mathbf{L} \{h(t)\}$$

For by definition,

$$\begin{aligned} \text{L.H.S.} &= \int_0^{\infty} e^{-st} [af(t) + bg(t) - ch(t)] dt \\ &= a \int_0^{\infty} e^{-st} f(t) dt + b \int_0^{\infty} e^{-st} g(t) dt - c \int_0^{\infty} e^{-st} h(t) dt = a\mathbf{L}\{f(t)\} + b\mathbf{L}\{g(t)\} - c\mathbf{L}\{h(t)\} \end{aligned}$$

This result can easily be generalised.

Because of the above property of L , it is called a *linear operator*.

II. First shifting property. If $\mathbf{L}\{f(t)\} = \bar{f}(s)$, then

$$\mathbf{L}\{e^{at}f(t)\} = \bar{f}(s-a).$$

$$\begin{aligned} \text{By definition, } \mathbf{L}\{e^{at}f(t)\} &= \int_0^{\infty} e^{-st} e^{at} f(t) dt = \int_0^{\infty} e^{-(s-a)t} f(t) dt \\ &= \int_0^{\infty} e^{-rt} f(t) dt, \text{ where } r = s-a = \bar{f}(r) = \bar{f}(s-a). \end{aligned}$$

Thus, if we know the transform $\bar{f}(s)$ of $f(t)$, we can write the transform of $e^{at}f(t)$ simply replacing s by $s-a$ to get $\bar{f}(s-a)$.

Application of this property leads us to the following useful results :

(1) $\mathbf{L}(e^{at}) = \frac{1}{s-a}$	$\left[\because \mathbf{L}(1) = \frac{1}{s} \right]$
(2) $\mathbf{L}(e^{at}t^n) = \frac{n!}{(s-a)^{n+1}}$	$\left[\because \mathbf{L}(t^n) = \frac{n!}{s^{n+1}} \right]$
(3) $\mathbf{L}(e^{at} \sin bt) = \frac{b}{(s-a)^2 + b^2}$	$\left[\because \mathbf{L}(\sin bt) = \frac{b}{s^2 + b^2} \right]$
(4) $\mathbf{L}(e^{at} \cos bt) = \frac{s-a}{(s-a)^2 + b^2}$	$\left[\because \mathbf{L}(\cos bt) = \frac{s}{s^2 + b^2} \right]$
(5) $\mathbf{L}(e^{at} \sinh bt) = \frac{b}{(s-a)^2 - b^2}$	$\left[\because \mathbf{L}(\sinh bt) = \frac{b}{s^2 - b^2} \right]$
(6) $\mathbf{L}(e^{at} \cosh bt) = \frac{s-a}{(s-a)^2 - b^2}$	$\left[\because \mathbf{L}(\cosh bt) = \frac{s}{s^2 - b^2} \right]$

where in each case $s > a$.

Example 21.1. Find the Laplace transforms of

(i) $\sin 2t \sin 3t$

(ii) $\cos^2 2t$

(iii) $\sin^3 2t$.

Solution. (i) Since $\sin 2t \sin 3t = \frac{1}{2} [\cos t - \cos 5t]$

$$\therefore \mathbf{L}(\sin 2t \sin 3t) = \frac{1}{2} [\mathbf{L}(\cos t) - \mathbf{L}(\cos 5t)] = \frac{1}{2} \left[\frac{s}{s^2 + 1^2} - \frac{s}{s^2 + 5^2} \right] = \frac{12s}{(s^2 + 1)(s^2 + 25)}$$

(ii) Since $\cos^2 2t = \frac{1}{2} (1 + \cos 4t)$

$$\therefore \mathbf{L}(\cos^2 2t) = \frac{1}{2} [\mathbf{L}(1) + \mathbf{L}(\cos 4t)] = \frac{1}{2} \left(\frac{1}{s} + \frac{s}{s^2 + 16} \right)$$

(iii) Since $\sin 6t = 3 \sin 2t - 4 \sin^3 2t$
 or $\sin^3 2t = \frac{3}{4} \sin 2t - \frac{1}{4} \sin 6t$
 $\therefore L(\sin^3 2t) = \frac{3}{4} L(\sin 2t) - \frac{1}{4} L(\sin 6t)$

$$= \frac{3}{4} \cdot \frac{2}{s^2 + 2^2} - \frac{1}{4} \cdot \frac{2}{s^2 + 6^2} = \frac{48}{(s^2 + 4)(s^2 + 36)}.$$

Example 21.2. Find the Laplace transform of

(i) $e^{-3t}(2 \cos 5t - 3 \sin 5t)$. (ii) $e^{2t} \cos^2 t$ (V.T.U., 2006) (iii) $\sqrt{t}e^{3t}$. (P.T.U., 2009)

Solution. (i) $L[e^{-3t}(2 \cos 5t - 3 \sin 5t)] = 2L(e^{-3t} \cos 5t) - 3L(e^{-3t} \sin 5t)$

$$= 2 \cdot \frac{s+3}{(s+3)^2 + 5^2} - 3 \cdot \frac{5}{(s+3)^2 + 5^2} = \frac{2s-9}{s^2 + 6s + 34}.$$

(ii) Since $L(\cos^2 t) = \frac{1}{2} L(1 + \cos 2t) = \frac{1}{2} \left\{ \frac{1}{s} + \frac{s}{s^2 + 4} \right\}$

$\therefore$ by shifting property, we get

$$L(e^{2t} \cos^2 t) = \frac{1}{2} \left\{ \frac{1}{s-2} + \frac{s-2}{(s-2)^2 + 4} \right\}.$$

(iii) Since $L(\sqrt{t}) = \frac{\Gamma(3/2)}{s^{3/2}} = \frac{(1/2) \cdot \Gamma\pi}{s^{3/2}}$

$\therefore$ by shifting property, we obtain $L(e^{3t} \sqrt{t}) = \frac{\sqrt{\pi}}{2} \frac{1}{(s-3)^{3/2}}.$

Example 21.3. If $L f(t) = \bar{f}(s)$, show that

$$L[(\sinh at) f(t)] = \frac{1}{2} [\bar{f}(s-a) - \bar{f}(s+a)]$$

$$L[(\cosh at) f(t)] = \frac{1}{2} [\bar{f}(s-a) + \bar{f}(s+a)]$$

Hence evaluate (i) $\sinh 2t \sin 3t$ (ii) $\cosh 3t \cos 2t$.

Solution. We have $L[(\sinh at) f(t)] = L\left[\frac{1}{2}(e^{at} - e^{-at}) f(t)\right] = \frac{1}{2} [L\{e^{at} f(t)\} - L\{e^{-at} f(t)\}]$

$$= \frac{1}{2} [\bar{f}(s-a) - \bar{f}(s+a)], \text{ by shifting property.}$$

Similarly, $L[(\cosh at) f(t)] = \frac{1}{2} [L\{e^{at} f(t)\} + L\{e^{-at} f(t)\}]$

$$= \frac{1}{2} [\bar{f}(s-a) + \bar{f}(s+a)], \text{ by shifting property.}$$

(i) Since $L(\sin 3t) = \frac{3}{s^2 + 3^2}$, the first result gives

$$L(\sinh 2t \sin 3t) = \frac{1}{2} \left\{ \frac{3}{(s-2)^2 + 3^2} - \frac{3}{(s+2)^2 + 3^2} \right\} = \frac{12s}{s^4 + 10s^2 + 169}$$

(ii) Since $L(\cos 2t) = \frac{s}{s^2 + 2^2}$, the second result gives

$$L(\cosh 3t \cos 2t) = \frac{1}{2} \left\{ \frac{s-3}{(s-3)^2 + 2^2} + \frac{s+3}{(s+3)^2 + 2^2} \right\} = \frac{2s(s^2 - 5)}{s^4 - 10s^2 + 169}.$$

Example 21.4. Show that

$$(i) L(t \sin at) = \frac{2as}{(s^2 + a^2)^2} \quad (\text{Bhopal, 2001}) \quad (ii) L(t \cos at) = \frac{s^2 - a^2}{(s^2 + a^2)^2}.$$

Solution. Since $L(t) = 1/s^2$. $\therefore L(te^{iat}) = \frac{1}{(s - ia)^2} = \frac{(s + ia)^2}{[(s - ia)(s + ia)]^2}$

or

$$L[t(\cos at + i \sin at)] = \frac{(s^2 - a^2) + i(2as)}{(s^2 + a^2)^2}$$

Equating the real and imaginary parts from both sides, we get the desired results.

Example 21.5. Find the Laplace transform of $f(t)$ defined as

$$(i) f(t) = t/\tau, \text{ when } 0 < t < \tau \\ = 1, \text{ when } t > \tau.$$

(Kerala, 2005)

$$(ii) f(t) = \begin{cases} 1, & 0 < t \leq 1 \\ t, & 1 < t \leq 2 \\ 0, & t > 2 \end{cases}$$

(J.N.T.U., 2006 ; W.B.T.U., 2005)

Solution. (i) $Lf(t) = \int_0^\tau e^{-st} \cdot \frac{t}{\tau} dt + \int_\tau^\infty e^{-st} \cdot 1 dt = \frac{1}{\tau} \left[t \cdot \frac{e^{-st}}{-s} \Big|_0^\tau - \int_0^\tau 1 \cdot \frac{e^{-st}}{-s} dt \right] + \left[\frac{e^{-st}}{-s} \Big|_\tau^\infty \right]$

$$= \frac{1}{\tau} \left[\frac{te^{-s\tau} - 0}{-s} - \left[\frac{e^{-st}}{s^2} \Big|_0^\tau \right] \right] + \frac{0 - e^{-s\tau}}{-s} = \frac{-e^{-s\tau}}{s} - \frac{e^{-s\tau} - 1}{\tau s^2} + \frac{e^{-s\tau}}{s} = \frac{1 - e^{-s\tau}}{\tau s^2}.$$

(ii) $L\{f(t)\} = \int_0^1 e^{-st} \cdot 1 dt + \int_1^2 e^{-st} \cdot t dt + \int_2^\infty e^{-st} \cdot (0) dt$

$$= \left[\frac{e^{-st}}{-s} \Big|_0^1 + \left[t \cdot \frac{e^{-st}}{-s} - \frac{e^{-st}}{s^2} \right]_1^2 \right] = \frac{1 - e^{-s}}{s} + \left\{ \left(-\frac{2e^{-2s}}{s} - \frac{e^{-2s}}{s^2} \right) - \left(-\frac{e^{-s}}{s} - \frac{e^{-s}}{s^2} \right) \right\}$$

$$= \frac{1}{s} - \frac{2e^{-2s}}{s} + \frac{e^{-s}}{s^2} - \frac{e^{-2s}}{s^2}.$$

Example 21.6. Find the Laplace transform of (i) $\left(\sqrt{t} - \frac{1}{\sqrt{t}}\right)^3$.

(Kurukshetra, 2005)

(ii) $\frac{\cos \sqrt{t}}{\sqrt{t}}$

(Mumbai, 2009)

Solution. (i) Since $(\sqrt{t} - 1/\sqrt{t})^3 = t^{3/2} - 3t^{1/2} + 3t^{-1/2} - t^{-3/2}$

$$\therefore L(\sqrt{t} - 1/\sqrt{t}) = L(t^{3/2}) - 3L(t^{1/2}) + 3L(t^{-1/2}) - L(t^{-3/2})$$

$$= \frac{\Gamma(3/2 + 1)}{s^{3/2+1}} - 3 \frac{\Gamma(1/2 + 1)}{s^{1/2+1}} + 3 \frac{\Gamma(-1/2 + 1)}{s^{-1/2+1}} - \frac{\Gamma(-3/2 + 1)}{s^{-3/2+1}}$$

$$= \frac{3}{2} \frac{\Gamma\left(\frac{3}{2}\right)}{s^{5/2}} - 3 \frac{\frac{1}{2} \Gamma\left(\frac{1}{2}\right)}{s^{3/2}} + 3 \frac{\Gamma\left(\frac{1}{2}\right)}{s^{1/2}} - \frac{\Gamma\left(-\frac{1}{2}\right)}{s^{-1/2}}$$

$$= \frac{3}{4} \frac{\sqrt{\pi}}{s^{5/2}} - \frac{3}{2} \frac{\sqrt{\pi}}{s^{3/2}} + \frac{3\sqrt{\pi}}{s^{1/2}} + \frac{2\sqrt{\pi}}{s^{-1/2}}$$

$$= \frac{\sqrt{\pi}}{4} \left(\frac{3}{s^{5/2}} - \frac{6}{s^{3/2}} + \frac{12}{s^{1/2}} + \frac{8}{s^{-1/2}} \right).$$

$$\left[\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi} \right]$$

(ii) We know that $\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots \infty$

$$\therefore \cos \sqrt{t} = 1 - \frac{t}{2!} + \frac{t^2}{4!} - \frac{t^3}{6!} + \dots$$

and

$$\frac{\cos \sqrt{t}}{\sqrt{t}} = t^{-1/2} - \frac{t^{1/2}}{2!} + \frac{t^{3/2}}{4!} - \frac{t^{5/2}}{6!} + \dots$$

and

$$\begin{aligned} L\left(\frac{\cos \sqrt{t}}{\sqrt{t}}\right) &= \frac{\Gamma(1/2)}{s^{1/2}} - \frac{1}{2!} \frac{\Gamma(3/2)}{s^{3/2}} + \frac{1}{4!} \frac{\Gamma(5/2)}{s^{5/2}} - \frac{1}{6!} \frac{\Gamma(7/2)}{s^{7/2}} + \dots \\ &= \frac{\Gamma(1/2)}{\sqrt{s}} - \frac{1}{2} \cdot \frac{1/2 \Gamma(1/2)}{s^{3/2}} + \frac{1}{4!} \frac{3/2 \cdot 1/2 \cdot \Gamma(1/2)}{s^{5/2}} - \frac{1}{6!} \frac{5/2 \cdot 3/2 \cdot 1/2 \cdot \Gamma(1/2)}{s^{7/2}} + \dots \\ &= \sqrt{\left(\frac{\pi}{2}\right)} \left[1 - \frac{1}{(4s)} + \frac{1}{2!} \frac{1}{(4s)^2} - \frac{1}{3!} \frac{1}{(4s)^3} \dots \right] = \sqrt{\left(\frac{\pi}{s}\right)} e^{-1/4s}. \end{aligned}$$

Example 21.7. Find the Laplace transform of the function

(i) $f(t) = |t-1| + |t+1|, t \geq 0$

(S.V.T.U., 2009)

(ii) $f(t) = [t]$, where $[]$ stands for the greatest integer function.

(P.T.U., 2010)

Solution. (i) Given function is equivalent to

$$f(t) = \begin{cases} 2, & 0 \leq t < 1 \\ 2t, & t \geq 1 \end{cases}$$

$$\begin{aligned} \therefore Lf(t) &= \int_0^1 e^{-st} (2) dt + \int_1^\infty e^{-st} (2t) dt = 2 \left[\left. \frac{e^{-st}}{-s} \right|_0^1 + 2 \left. \left(\frac{t e^{-st}}{-s} \right) \right|_1^\infty - \left. \frac{e^{-st}}{(-s)^2} \right|_1^\infty \right] \\ &= 2 \left(\frac{e^{-s}}{-s} + \frac{1}{s} \right) + 2 \left(\frac{0 - e^{-s}}{-s} - \frac{0 - e^{-s}}{s^2} \right) = \frac{2}{s} \left(1 + \frac{e^{-s}}{s} \right) \end{aligned}$$

(ii) Given function is equivalent to

$$[t] = 0 \text{ in } (0, 1) + 1 \text{ in } (1, 2) + 2 \text{ in } (2, 3) + 3 \text{ in } (3, 4) + \dots$$

$$\begin{aligned} \therefore L[f(t)] &= \int_0^\infty e^{-st} [f(t)] dt = \int_0^\infty e^{-st} [t] dt \\ &= \int_0^1 e^{-st} (0) dt + \int_1^2 e^{-st} (1) dt + \int_2^3 e^{-st} (2) dt + \int_3^4 e^{-st} (3) dt + \dots \infty \\ &= 0 + \left. \frac{e^{-st}}{-s} \right|_1^2 + 2 \left. \frac{e^{-st}}{-s} \right|_2^3 + 3 \left. \frac{e^{-st}}{-s} \right|_3^4 + \dots \infty \\ &= -\frac{1}{s} [(e^{-2s} - e^{-s}) + 2(e^{-3s} - e^{-2s}) + 3(e^{-4s} - e^{-3s}) + \dots \infty] \\ &= \frac{1}{s} (e^{-s} + e^{-2s} + e^{-3s} + \dots \infty) = \frac{1}{s} \left(\frac{e^{-s}}{1 - e^{-s}} \right) = \frac{1}{s(e^s - 1)}. \end{aligned}$$

III. Change of scale property. If $L\{f(t)\} = \bar{f}(s)$, then $L\{f(at)\} = \frac{1}{a} \bar{f}\left(\frac{s}{a}\right)$

$$\begin{aligned} L\{f(at)\} &= \int_0^\infty e^{-st} f(at) dt = \int_0^\infty e^{-su/a} f(u) \cdot du/a \\ &= \frac{1}{a} \int_0^\infty e^{-su/a} f(u) du = \frac{1}{a} \bar{f}(s/a). \end{aligned}$$

$$\left| \begin{array}{l} \text{Put } at = u \\ dt = du/a \end{array} \right.$$

Example 21.8. Find $L\left\{\frac{\sin at}{t}\right\}$, given that $L\left\{\frac{\sin t}{t}\right\} = \tan^{-1}\left\{\frac{1}{s}\right\}$.

Solution. By the above property,

$$L\left\{\frac{\sin at}{at}\right\} = \frac{1}{a} \tan^{-1}\left\{\frac{1}{(s/a)}\right\} = \frac{1}{a} \tan^{-1}\left(\frac{a}{s}\right) \text{ i.e., } L\left\{\frac{\sin at}{t}\right\} = \tan^{-1}\left\{\frac{a}{s}\right\}.$$

PROBLEMS 21.1

Find the Laplace transforms of

1. $e^{2t} + 4t^3 - 2 \sin 3t + 3 \cos 3t$. (J.N.T.U., 2003)
2. $1 + 2\sqrt{t} + 3/\sqrt{t}$.
3. $3 \cosh 5t - 4 \sinh 5t$. (Nagarjuna, 2006)
4. $\cos(at + b)$.
5. $(\sin t - \cos t)^2$.
6. $\sin 2t \cos 3t$. (Kottayam, 2005)
7. $\sin \sqrt{t}$.
8. $\sin^5 t$. (Mumbai, 2007)
9. $\cos^3 2t$.
10. $e^{-at} \sinh bt$.
11. $e^{2t}(3t^5 - \cos 4t)$. (P.T.U., 2007)
12. $e^{-3t} \sin 5t \sin 3t$. (V.T.U., 2006)
13. $e^{-t} \sin^2 t$. (Mumbai, 2009)
14. $e^{2t} \sin^4 t$. (Mumbai, 2007)
15. $\cosh at \sin at$. (Delhi, 2002)
16. $\sinh 3t \cos^2 t$. (Madras, 2000)
17. $t^2 e^{2t}$. (V.T.U., 2008 S)
18. $(1 + te^{-t})^3$.
19. $t \sqrt{1 + \sin t}$. (Mumbai, 2007)
20. $f(t) = \begin{cases} 4, & 0 \leq t \leq 1 \\ 3, & t > 1 \end{cases}$ (U.P.T.U., 2009)
21. $f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 0, & t > \pi. \end{cases}$ (Madras, 2000 S)
22. $f(x) = \begin{cases} \sin(x - \pi/3), & x > \pi/3 \\ 0, & x < \pi/3 \end{cases}$ (Rajasthan, 2006)
23. $f(t) = \begin{cases} \cos(t - 2\pi/3), & t > 2\pi/3 \\ 0, & t < 2\pi/3 \end{cases}$
24. $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ t - 1, & 2 < t < 3 \\ 7, & t > 3. \end{cases}$ (Mumbai, 2007)
25. If $L[f(t)] = \frac{1}{s(s^2 + 1)}$, find $L[e^{-t}f(2t)]$.

21.5 TRANSFORMS OF PERIODIC FUNCTIONS

If $f(t)$ is a periodic function with period T , i.e., $f(t + T) = f(t)$, then

$$L\{f(t)\} = \frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}$$

We have $L\{f(t)\} = \int_0^\infty e^{-st} f(t) dt = \int_0^T e^{-st} f(t) dt + \int_T^{2T} e^{-st} f(t) dt + \int_{2T}^{3T} e^{-st} f(t) dt + \dots$

In the second integral put $t = u + T$, in the third integral put $t = u + 2T$, and so on. Then

$$\begin{aligned} L\{f(t)\} &= \int_0^T e^{-st} f(t) dt + \int_0^T e^{-s(u+T)} f(u+T) du + \int_0^T e^{-s(u+2T)} f(u+2T) du + \dots \\ &= \int_0^T e^{-st} f(t) dt + e^{-sT} \int_0^T e^{-su} f(u) du + e^{-2sT} \int_0^T e^{-su} f(u) du + \dots \\ &\quad [\because f(u) = f(u+T) = f(u+2T) \text{ etc.}] \\ &= \int_0^T e^{-st} f(t) dt + e^{-sT} \int_0^T e^{-st} f(t) dt + e^{-2sT} \int_0^T e^{-st} f(t) dt + \dots \\ &= (1 + e^{-sT} + e^{-2sT} + \dots \infty) \int_0^T e^{-st} f(t) dt \\ &= \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt. \end{aligned}$$

(V.T.U., 2008 ; Mumbai, 2006)